

Visual Assets & Screenshot Attachment Guide

Complete Point-by-Point Blueprint for Documentation & Submission Slides

Purpose: This guide specifies exactly which diagrams, UI screenshots, architectural schemas, and user flow illustrations must be inserted into each of the 16 sections of the **AI Teacher Technical Documentation**. Each entry maps to an exact section number, provides the recommended figure title, describes what screen or diagram to capture, specifies placement within the text, and provides ready-to-use professional captions.

Standard	Recommended Guideline for Documentation & Submission Slides
Resolution	Capture desktop screenshots at 1920x1080 (1080p) or higher. Save as lossless PNG to maintain crystal-clear text readability.
Theme	Capture all screens in the default Dark Theme (`#0A0E1A`) for aesthetic consistency with the React UI.
Annotations	Use subtle indigo/violet highlight boxes or numbered badges (①, ②, ③) to highlight interactive buttons like the mic, time selector, and language switcher.
Diagram Tooling	For architecture and state machine diagrams (Sections 4, 6, 7), export from Mermaid.js, Draw.io, or Excalidraw with transparent backgrounds.
Captioning	Follow IEEE/ACM caption format: <i>Figure X.Y: Concise title explaining the pedagogical interaction shown.</i>

Master Visual Attachment Blueprint (Sections 1 to 16)

Point / Section	Image Title	What to Capture / Illustrate	Placement & Caption
1. Problem Statement	Figure 1.1: Digital Learning Modality Gap	Comparison chart: Pre-recorded static video vs Text-only chatbots vs AI Virtual Human Educator (showing passive vs active learning).	Placement: After para 2. Caption: <i>Fig 1.1: Cognitive limitations of static video & text AI compared to adaptive video tutoring.</i>
2. Solution Overview	Figure 2.1: End-to-End AI Teacher Pipeline	Flowchart: Ingestion → RAG Vector Store → Pedagogy State Machine → Multimodal Synthesis (Voice+Visual+Avatar) → Classroom UI.	Placement: Under Solution Overview. Caption: <i>Fig 2.1: Complete architectural flow from course materials to interactive video delivery.</i>
3. Key Features (A)	Figure 3.1: Dual Ingestion & Time Mode UI	Screenshot of 'Source & Topic' tab showing Direct Topic mode, Document Dropzone, and Adaptive Time pills (5m/20m/60m/7d).	Placement: Beside Ingestion row. Caption: <i>Fig 3.1: Source ingestion interface with personalized time-budgeting selectors.</i>
3. Key Features (B)	Figure 3.2: Interactive Video Classroom	Screenshot of 'Interactive Lesson' tab showing side-by-side Whiteboard visual diagram and talking AI teacher video.	Placement: Beside Classroom row. Caption: <i>Fig 3.2: 2-column multimodal classroom featuring whiteboard schemas and teacher video.</i>
3. Key Features (C)	Figure 3.3: Active Recall Flashcards & Notes	Screenshot of the 3D flip card modal showing front conceptual question, back analogy, and exported Markdown notes snippet.	Placement: Under Retention row. Caption: <i>Fig 3.3: Interactive active recall flashcards modal and exported study notes guide.</i>
4. System Architecture	Figure 4.1: Component System Architecture	Full architectural diagram showing React Vite SPA, FastAPI backend, SQLite, ChromaDB, Edge-TTS, and Groq Cloud APIs.	Placement: Directly below Section 4 heading. Caption: <i>Fig 4.1: Decoupled multi-tier system architecture of AI Teacher platform.</i>
5. AI/ML Models Used	Figure 5.1: Multimodal AI Model Stack	Block diagram highlighting Llama 3.3 70B (planning), Llama 3.1 8B (fast eval), Whisper Large v3 (STT), and Edge-TTS (voices).	Placement: Below models table. Caption: <i>Fig 5.1: Layered AI/ML model orchestration and fallback hierarchy.</i>
6. RAG Implementation	Figure 6.1: TOC-Aware Vector Indexing Pipeline	Diagram illustrating textbook PDF ingestion, Table of Contents page locator (e.g. Chapter 2 -> Page 41), chunking, and ChromaDB retrieval.	Placement: Under RAG section. Caption: <i>Fig 6.1: Table of Contents aware vector chunking and strict retrieval grounding pipeline.</i>
7. Prompt / Agent Architecture	Figure 7.1: Pedagogical Finite State Machine	State transition diagram: INIT → PLANNING → TEACHING → AWAITING_CHECKPOINT → REMEDIATION → ASSESSMENT → REPORT.	Placement: Below FSM description. Caption: <i>Fig 7.1: Finite State Machine state transitions governing pedagogical pacing.</i>
8. Personalization Approach	Figure 8.1: AI Teacher Persona Trio	Side-by-side graphic of Dr. Sarah (Intuitive), Prof. Aryan (Analytical), and Coach Maya (Exam-Prep) with persona tone descriptions.	Placement: Beside Personas subsection. Caption: <i>Fig 8.1: Selectable teacher personalities with custom avatars, vocal timbres, and pedagogical styles.</i>
9. Assessment Methodology	Figure 9.1: Checkpoint & Remediation Loop	Screenshot showing student answering a checkpoint question, the evaluation rubric feedback, and the simplified remediation branch.	Placement: Under Formative Assessment. Caption: <i>Fig 9.1: Formative micro-checkpoint with live rubric scoring and remediation branch.</i>
9. Assessment Methodology	Figure 9.2: Diagnostic Learning Report	Screenshot of the final assessment report displaying Bloom's taxonomy radar chart, score breakdown, strengths, and weaknesses.	Placement: Under Summative Assessment. Caption: <i>Fig 9.2: Comprehensive student diagnostic report and revision roadmap.</i>

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10. Multilingual Implementation	Figure 10.1: In-Lesson Language Switcher	Screenshot showing the [English Hindi Hinglish] pills on the video player with Hindi script and Devanagari whiteboard subtitles.	Placement: Under Multilingual section. Caption: Fig 10.1: Dynamic in-lesson multilingual switcher re-narrating the lesson into Hindi/Hinglish.
11. Voice Implementation	Figure 11.1: Audio Narration & STT Waveform	Graphic showing dual voice pipeline: Edge-TTS neural speech generation + Web Speech API zero-latency microphone speech recognition.	Placement: Beside Voice section. Caption: Fig 11.1: Bidirectional voice architecture: Neural Edge-TTS output & Web Speech STT input.
12. Avatar / Video Generation	Figure 12.1: Real-Time Lip-Sync Synthesis	Visual diagram of the 4-step OpenCV video pipeline: Audio envelope extraction → Face landmark detection → Mouth warp → MP4 encoding in <2s.	Placement: Under Video Generation. Caption: Fig 12.1: Sub-2-second audio-driven facial motion & lip-sync synthesis architecture.
13. APIs & Third-Party Services	Figure 13.1: Cloud Services Integration	Service topology diagram connecting Groq Cloud, Edge-TTS, LangSmith, Pollinations AI, and ChromaDB.	Placement: Under APIs section. Caption: Fig 13.1: External API topology and asynchronous communication channels.
14. Setup Instructions	Figure 14.1: Full-Stack Launch Terminal	Screenshot of terminal running 'run_fullstack.bat' showing FastAPI backend (port 8000) and Vite React frontend (port 5173).	Placement: Below code block. Caption: Fig 14.1: Automated full-stack execution with separate backend and frontend runners.
15. Deployment Instructions	Figure 15.1: Cloud Deployment on Render	Screenshot of Render.com dashboard displaying multi-stage Docker build logs, green 'Live' status, and public HTTPS domain URL.	Placement: Under Deployment section. Caption: Fig 15.1: Render.com production deployment dashboard with active HTTPS service.
16. Known Limitations	Figure 16.1: Technical Trade-Off Matrix	Trade-off matrix diagram comparing 2D OpenCV affine lip-sync vs 3D NeRF compute overhead, and SQLite vs PostgreSQL clustering.	Placement: At end of section 16. Caption: Fig 16.1: Performance trade-off matrix: sub-2-second latency vs heavy 3D mesh rendering.

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